RISHOBAN KANDEEPAN

Embedded Firmware Development | Embedded C/C++ | ARM Cortex-M/ESP32 | RTOS & Secure Systems

Colombo, Sri Lanka | rishobankandeepan@gmail.com | +9475 768 19 94

linkedin.com/in/rishoban-kandeepan | Rishoban Kandeepan-Portfolio | github.com/RishobanK

Summary

Embedded firmware developer with hands-on experience in Embedded C/C++, STM32/ARM Cortex-M, ESP32, microcontroller-based control systems, sensor integration, and hardware-software debugging. Experienced in bootloader development, firmware validation, flash memory handling, CRC32 integrity checking, AGV firmware development, UART/I2C/SPI communication, GPIO, PWM, ADC, EEPROM storage, and real-time sensor handling. Strong interest in secure embedded systems, firmware update mechanisms, authenticated communication, and reliable firmware design. Currently strengthening knowledge in FreeRTOS, Embedded Linux, secure firmware updates, firmware signing, and authenticated embedded communication.

Technical Skills

Languages: C, C++, Python, Assembly; basic Java

Firmware & Low-Level: Bare-metal firmware, bootloader design, flash programming, linker scripts, vector table relocation, interrupts, timers/PWM, ADC, EEPROM/NVM, state machines, serial debugging

Platforms & Interfaces: STM32/ARM Cortex-M, ESP32, ATmega2560, PIC16F877A, Raspberry Pi; UART, I2C, SPI, GPIO, PWM, ADC, TCP/IP

RTOS, Linux & Security: FreeRTOS concepts, Embedded Linux basics, firmware validation, CRC32 integrity checking, secure boot concepts, firmware signing, TLS/HTTPS, certificate validation

Tools & PCB Design: Git, GitHub, PlatformIO, STM32CubeIDE, STM32CubeMX, ST-Link, KiCad, EasyEDA, Proteus

Certifications & Professional Development

Selected Certifications: Advanced BSP Development with Embedded C; Embedded Software Architecture; Embedded Software Design; Embedded Systems using C.

Current Focus: FreeRTOS, Embedded Linux, secure firmware updates, firmware signing, secure boot concepts, TLS/HTTPS, certificate validation, and authenticated embedded communication.

Experience

Electronic Engineering Trainee, Variosystems – Sri Lanka June 2025 – Nov 2025

- Developed and tested embedded firmware improvements for an industrial AGV, including load-cell calibration, EEPROM persistence, PID line-following, white-zone detection, adaptive turning, non-blocking timer logic, sensor calibration, route validation, and hardware-software debugging.
- Developed a Raspberry Pi-based PCB alignment verification system using Python, Tkinter, and OpenCV, with conveyor/buzzer outputs triggered by real-time orientation validation.

Engineering Trainee, Colombo Dockyard PLC – Sri Lanka Jan 2023 – April 2023

- Assisted in maintenance and troubleshooting of shipboard electrical and communication systems, supporting operational reliability and safety.
- Contributed to energy-efficiency improvements for welding machines by identifying idle power usage issues and supporting practical optimization work.

Education

Sri Lanka Institute of Information Technology (SLIIT), BSc (Hons) in Electrical & Electronic Engineering June 2021 – Present

- **Relevant Coursework:** Embedded Systems Programming, Digital Design, Electronic Design, Robotics, IoT, Computer Vision

University of Moratuwa, Bachelor of Information Technology (BIT) – First semester Feb 2026 – Present

Royal College, Colombo, G.C.E. Advanced Level and Ordinary Level completed Jan 2012 – Oct 2020

Projects

Secure STM32 Bootloader & Firmware Update System – ARM Cortex-M Firmware Security GitHub

- Developed a custom STM32 Blue Pill bootloader with relocated application execution, flash erase/write handling, update flag mechanism, and firmware update workflow.
- Verified bootloader behavior using LED-coded status indicators, ST-Link breakpoints, watch variables, STM32CubeProgrammer flash memory inspection, and ESP32-C3 UART bridge testing.
- **Tools:** STM32 Blue Pill, Embedded C, STM32CubeIDE, ST-Link, Flash Memory, Linker Script, CRC32, Python

STM32F103 Secure Telemetry Node with FreeRTOS – Secure Embedded Communication GitHub

- Designing a FreeRTOS-based STM32F103 telemetry application using Embedded C, USART1 communication, task-based firmware structure, and PC13 GPIO control.
- Planned features include UART packet parsing, checksum/CRC validation, token-based command authentication, telemetry reporting, and LED/status task handling.
- **Tools:** STM32F103C8T6 Blue Pill, FreeRTOS, Embedded C, STM32CubeIDE, USART1, GPIO, ST-Link, Git/GitHub

Autonomous Guided Vehicle (AGV) – Embedded Firmware & Sensor Integration GitHub

- Developed embedded firmware for an industrial AGV with load-cell calibration, EEPROM persistence, PID-based line following, white-zone detection, adaptive turning, pushbutton override, OLED feedback, and non-blocking timer logic.
- Performed sensor calibration, route validation, and hardware-software debugging to improve navigation reliability under industrial operating conditions.
- **Tools:** Embedded C/C++, ATmega2560, CZL601 Load Cell, IR Sensor Array, EEPROM, OLED, Motor Control

ESP32 Custom Development Board – PCB Design & Embedded Hardware GitHub

- Designed a custom ESP32-based development board for embedded prototyping, including power regulation, boot/reset circuitry, GPIO breakout, programming interface, and sensor/IO expansion.
- Created schematic, PCB layout, and fabrication files while considering component placement, signal routing, power distribution, and debugging accessibility.
- **Tools:** ESP32, KiCad, PCB Design, Schematic Design, Power Regulation, GPIO, UART/I2C/SPI

Wearable Kyphosis Monitoring System – ESP32 Sensor Acquisition & Feedback GitHub

- Developed an ESP32-based wearable monitoring system using EMG and gyroscope sensors for real-time posture-related data acquisition and feedback control.
- Implemented sensor data reading, noise filtering, serial data output, and vibration feedback using micro DC vibration motors.
- **Tools:** ESP32, Raspberry Pi 5, MyoWare Sensor, MPU6050, Embedded C/C++, Python

Vision-Based PCB Alignment Verification – Raspberry Pi Automation GitHub

- Developed a Raspberry Pi-based PCB alignment verification system using Python, OpenCV, Tkinter, ORB feature matching, and real-time ROI selection.
- Integrated orientation validation with conveyor/buzzer outputs to support automated inspection under rotation, scale variation, and partial occlusion.
- **Tools:** Raspberry Pi 5, Python, OpenCV, Tkinter, GPIO

Volunteering

Robofest 2023 & 2024 – Supported outreach activities for a nationwide robotics competition, including student guidance on line-following robot concepts.

Referees

Dr. Eranga Wijesinghe
Professor, Faculty of Engineering, SLIIT
eranga.w@sliit.lk | +9474 153 10 13

Eng. T. Shanmugam
Director, Interprof International
s_thangamuthu@yahoo.com | +9477 066 28 25